

Multiple Choice (10 marks)

1. If the frequency of an oscillating source on the surface of a pool of water is 3 Hz, what is the speed of the wave if the wavelength is observed to be 0.5 m?
 - (a) 0.75 m/sec
 - (b) 3.5 m/sec
 - (c) 0.5 m/sec
 - (d) 1.5 m/sec
 - (e) 6 m/sec
2. Find the index of refraction for a glass whose dielectric constant is 2.5.
 - (a) 2.25
 - (b) 2.0
 - (c) 2.5
 - (d) 1.41
 - (e) 0.5
3. How many electrons would have to be removed from a coin to leave it with a charge of $+1.0 \times 10^{-7}\text{C}$?
 - (a) 8.5×10^{11}
 - (b) 7.1×10^{11}
 - (c) 7.5×10^{11}
 - (d) 9.2×10^{11}
 - (e) 3.8×10^{11}
4. When an increase in speed doubles the momentum of a moving body its kinetic energy
 - (a) decreases slightly
 - (b) increases but does not reach double
 - (c) remains the same
 - (d) halves
 - (e) quadruples
5. The amount of air drag on an 0.8-N flying squirrel dropping vertically at terminal velocity is
 - (a) dependent on the speed of the wind

- (b) greater than 0.8 N
- (c) 0.8 N
- (d) dependent on the temperature of the air
- (e) dependent on the orientation of its body

True / False (10 marks)

Answer True or False

- 6. True or False: The power factor of the circuit can exceed 1, indicating an efficient energy transfer.
- 7. True or False: The semi-major axis of the asteroid Pallas's orbit is 4.15×10^{11} m.
- 8. True or False: The force acting on a 1000-gram mass sliding down an inclined plane is 750,000 dynes.
- 9. True or False: The energy equivalent of a particle with a mass of 1 amu is approximately 931 MeV.
- 10. True or False: The force of the Earth on the elevator is the action-reaction force pair to the man's weight in the upward-moving elevator.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Analyze a one-dimensional inelastic collision between a particle of mass $2m$ and a stationary particle of mass m . Discuss the fraction of initial kinetic energy lost during the collision and the underlying physics principles.
- 12. Discuss the relationship between frequency and wavelength in sound waves, and calculate the wavelength of the sound wave produced by a standard tuning fork vibrating at 440 Hz.

End of Paper